Integration by Substitution (U-Substitution)

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: _____ Date: _____

Learning Objectives

- Identify the appropriate substitution (u) for a composite integrand
- Rewrite an integral entirely in terms of u and du, adjusting for constant factors
- Integrate using the Power Rule and back-substitute to express the answer in terms of x

Problems

1. Use the substitution rule to find the integral of 3x squared times the square root of x cubed plus 5, with respect to x. Let u equal x cubed plus 5.

$$\int 3x^2\sqrt{x^3+5}\,dx$$

2. Find the integral of 4x times the quantity x squared plus 7, raised to the third power, with respect to x. Let u equal x squared plus 7.

$$\int 4x(x^2+7)^3\,dx$$

3. Use u-substitution to evaluate the integral of 5x to the fourth power times the quantity x to the fifth minus 2, raised to the fourth power, with respect to x.

$$\int 5x^4(x^5-2)^4\,dx$$

4. Find the integral of x cubed times the quantity x to the fourth plus 1, raised to the fifth power, with respect to x. Note that you will need to adjust for a constant factor.

$$\int x^3(x^4+1)^5\,dx$$

5. Use u-substitution to evaluate the integral of secant squared of 3x, with respect to x.

$$\int \sec^2(3x)\,dx$$

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6. Find the integral of secant squared of the quantity 1 over x, all divided by x squared, with respect to x. Let u equal 1 over x.

7. Evaluate the integral of 6x to the fifth power divided by the square root of x to the sixth plus 4, with respect to x.

$$\int \frac{6x^5}{\sqrt{x^6+4}}\,dx$$

8. Use u-substitution to find the integral of x squared times the quantity 2 plus x cubed, raised to the negative second power, with respect to x.

$$\int x^2(2+x^3)^{-2}\,dx$$

9. Evaluate the integral of the quantity 3x squared minus 2 times the quantity x cubed minus 2x, raised to the seventh power, with respect to x.

$$\int (3x^2-2)(x^3-2x)^7\,dx$$

10. Find the integral of x squared times the square root of the quantity 4 minus x cubed, raised to the fifth power, with respect to x. (Hint: rewrite the radical as a fractional exponent and adjust for constant factors.)

$$\int x^2(4-x^3)^{5/2}\,dx$$

Integration by Substitution (U-Substitution) — Answer Key

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Answer Key

1. Answer: (2/3)(x³ + 5)^(3/2) + C

- Let u = x³ + 5, so du = 3x² dx
- Rewrite: ∫ √u du
- Rewrite as: ∫ u^(1/2) du
- Integrate: u^(3/2) / (3/2) + C = (2/3)u^(3/2) + C
- Back-substitute: (2/3)(x³ + 5)^(3/2) + C

2. Answer: (x² + 7)⁴ / 2 + C

- Let u = x² + 7, so du = 2x dx, meaning 4x dx = 2 du
- Rewrite: ∫ 2u³ du
- Integrate: 2 · u⁴/4 + C = u⁴/2 + C
- Back-substitute: (x² + 7)⁴ / 2 + C

3. Answer: (x⁵ – 2)⁵ / 5 + C

- Let u = x⁵ – 2, so du = 5x⁴ dx
- Rewrite: ∫ u⁴ du
- Integrate: u⁵/5 + C
- Back-substitute: (x⁵ – 2)⁵ / 5 + C

4. Answer: (x⁴ + 1)⁶ / 24 + C

- Let u = x⁴ + 1, so du = 4x³ dx, meaning x³ dx = (1/4) du
- Rewrite: ∫ (1/4) u⁵ du
- Integrate: (1/4) · u⁶/6 + C = u⁶/24 + C
- Back-substitute: (x⁴ + 1)⁶ / 24 + C

5. Answer: (1/3) tan(3x) + C

- Let u = 3x, so du = 3 dx, meaning dx = (1/3) du
- Rewrite: ∫ (1/3) sec²(u) du
- Integrate: (1/3) tan(u) + C
- Back-substitute: (1/3) tan(3x) + C

6. Answer: –tan(1/x) + C

- Let u = 1/x = x^(–1), so du = –x^(–2) dx = –(1/x²) dx
- Therefore (1/x²) dx = –du
- Rewrite: ∫ sec²(u) · (–du) = –∫ sec²(u) du
- Integrate: –tan(u) + C
- Back-substitute: –tan(1/x) + C

7. Answer: $2\sqrt{(x^6 + 4)} + C$

- Let $u = x^6 + 4$, so $du = 6x^5\,dx$
- Rewrite: $\int u^{(-1/2)}\,du$
- Integrate: $u^{(1/2)} / (1/2) + C = 2u^{(1/2)} + C$
- Back-substitute: $2\sqrt{(x^6 + 4)} + C$

8. Answer: $-1 / (3(2 + x^3)) + C$

- Let $u = 2 + x^3$, so $du = 3x^2\,dx$, meaning $x^2\,dx = (1/3)\,du$
- Rewrite: $\int (1/3)\,u^{(-2)}\,du$
- Integrate: $(1/3) \cdot u^{(-1)}/(-1) + C = -1/(3u) + C$
- Back-substitute: $-1 / (3(2 + x^3)) + C$

9. Answer: $(x^3 - 2x)^8 / 8 + C$

- Let $u = x^3 - 2x$, so $du = (3x^2 - 2)\,dx$
- Rewrite: $\int u^7\,du$
- Integrate: $u^8/8 + C$
- Back-substitute: $(x^3 - 2x)^8 / 8 + C$

10. Answer: $-2(4 - x^3)^{(7/2)} / 21 + C$

- Let $u = 4 - x^3$, so $du = -3x^2\,dx$, meaning $x^2\,dx = -(1/3)\,du$
- Rewrite: $\int u^{(5/2)} \cdot (-1/3)\,du = -(1/3) \int u^{(5/2)}\,du$
- Integrate: $-(1/3) \cdot u^{(7/2)} / (7/2) + C = -(1/3) \cdot (2/7)\,u^{(7/2)} + C$
- Simplify: $-2u^{(7/2)}/21 + C$
- Back-substitute: $-2(4 - x^3)^{(7/2)} / 21 + C$